**Q.1**

In a $3 \times 3$ matrix, how many minors can be calculated?

A. 3
B. 9
C. **12**
D. 6

---

**Q.2 Two matrices A and B are multiplied to get BA if**

A. both are rectangular
B. both have same order
C. number of rows of A is equal to number of columns of B
D. **number of columns of A is equal to number of columns of B**

---

**Q.3 For a geometric sequence, if 2nd term is equal to 4th term then value of common ratio is**

A. 1 only
B. **+1 or -1**
C. 0
D. -1 only

---

**Q.4**

For a $3 \times 3$ matrix, swapping two rows will:

A. Leave the determinant unchanged.
B. Double the determinant.
C. **Square the determinant.**
D. Change the sign of the determinant.

---

**Q.5 Cramer`s rule is not suitable for which type of problems?**

A. Systems with 3 unknowns
B. **Large systems**
C. Systems with two unknowns
D. Small systems with 4 unknowns

---

**Q.6**

Find the minor of the element $a_{11}$ in the matrix $\begin{bmatrix} 2 & 3 \\ 4 & 5 \end{bmatrix}$.

A. 2
B. 4
C. **1**
D. 5

---

**Q.7 What is the 31st term of the arithmetic sequence: 1, 4, 7, 10, …?**

A. 90
B. 93**C. 91**
D. 92

---

**Q.8 {\rm{The Cofactor of}}\,\,\,\,a_{ij} = \,\, - - - - .\,\,\,\,(Minor\,\,a_{ij} )**

**A. ( { - 1} )^{i + j}**
**B.** None\,\,of\,\,these
**C.** ( { - 1} )^{i  j} \,
**D.** ( { - 1} )^{i - j} \,

---

**Q.9 What is the next term in the sequence 1, 2, 4, 8, 16, _____ ?**

A. 36
B. **32**
C. 30
D. 24

---

**Q.10 If the first term=a=1, common ratio=r=1/2; then which term of geometric sequence is 1/4?**

A. 4
B. 6
C. **3**
D. 5

---

**Q.1 {\rm{The Cofactor of}}\,\,\,\,a_{ij} = \,\, - - - - .\,\,\,\,(Minor\,\,a_{ij} )**

**A. ( { - 1} )^{i + j}**

**B.** ( { - 1} )^{i  j} \,

**C.** None\,\,of\,\,these

**D.** ( { - 1} )^{i - j} \,

---

**Q.2 What is the next term in the sequence 1, 4, 9, 16, 25, _____ ?**

A. 40

B. 35**C. 36**

D. 30

---

**Q.3 The explicit formula when a1 = 6 and d = 2 is _____.**

**A. an = 6 + 2(n – 1)**

**B.** an = 6 + 2(n + 1)

**C.** an = 6 - 2(n – 1)D. an = 6 + 2n

---

**Q.4 What is the term after the next term in the sequence 5, 10, 15, 20, _____?**

A. 25

B. 35

C. 40

D. **30**

---

**Q.5 A Square matrix with ones on the diagonal and zeros elsewhere is called a/an ..... matrix**

**A. identity**

**B.** diagonal

**C.** Null

**D.** scalar

---

**Q.6 Given a square matrix A, if there exists a matrix B such that AB = BA = I, then A is called the ..... of B**

A. Additive inverse

B. **Multiplicative inverse**

C. Transpose

D. Adjoint

---

**Q.7 The first term of the sequence whose** A. 4

**“nth term = 2n; n>2 “ is-----.**

**B. 6**

**C.** 1

**D.** 2

---

**Q.8 Two matrices of same order can be.....**

A. subtracted but not added
B. added but not subtracted
C. multiplied
D. **added and subtracted**

**Q.9 Cramer`s rule is not suitable for which type of problems?**

A. Small systems with 4 unknowns
B. **Large systems**
C. Systems with two unknowns
D. Systems with 3 unknowns

---

**Q.10 What is the common difference in the arithmetic sequence: 7, 11, 15, 19, ...?**

**A. 4**
**B.** 5
**C.** 2
**D.** 3

**Q.1 Domain of function: f(x)=1/(Sqrt(x-1)) is all Real numbers ---------.**

A. x<1
B. **x>1**
C. except x=1
D. except x=0

---

**Q.2 If A = {1, 2, 3} and B = {2, 3, 4} then ……**

A. Intersection of A and B has 3 elements
B. B is subset of A
C. A is subset of B
D. **Union of A and B has 4 elements**

---

**Q.3 If f(x)=x and g(x)=1-x, then (f+g)(x) = ----------**

A. 1-2x
B. 2x
C. **1**
D. 2x-1

---

**Q.4 If f(x)=x and g(x)=1-x, then (f/g)(x) = ---------.**

A. 1
B. 1-(1/x)
C. (1-x)/x
D. **x/(1-x)**

---

**Q.5 Inverse of the function f(x)=3 doesn't exist due to which of the following reason?**

**A. It is many to one function**
**B.** It is 1-1 function
**C.** It is Bijective function
**D.** It is Odd function

---

**Q.6 If f(x)=x and g(x)=1-x, then (f-g)(x) = ----------**

A. 1-2x
B. **2x-1**
C. 2xD. -1

---

**Q.7 Which of these sets are equal {a,b,c}, {a,b,c,a},{a,b,b,c} and {a,c,b,c}?**

A. 2nd and 3rd only
B. **All of these**
C. None of these
D. 1st and 4th only

---

**Q.8 If\,f(x) = {x^2}\,and\,\,g(x) = \frac{1}{x},\,\,then\,\,fog(x) =**

A. {1}{x}
B. **{1}{{{x^2}}}**
C. x
D. none

---

**Q.9 Range of the function: f(x)=(1/x) is ---------.**

A. All Real numbers except 1
B. **All Real numbers except 0**
C. All Real numbers
D. All Real numbers except -1

---

**Q.10 2 is a**

**A. rational number**
**B.** natural number
**C.** irrational number
**D.** integer

---

**Q.1 Every real number is -------.**

A. a rational number
B. a positive integer
C. **a complex number**
D. a negative integer

---

**Q.2 Which of the following is the real part of complex number: z = i?**

A. 1
B. **0**
C. -1
D. None of these

---

**Q.3 Domain of the function:f(x)=1/(1-x) is ---------.**

A. All Real numbers
B. All Real numbers except -1
C. All Real numbers except 0
D. **All Real numbers except 1**

---

**Q.4 \,{\mathop{\rm Re}\nolimits} al\,\,part\,\,of\,\,3ab\left( {i + i^2 } \right)\,\,is\,\, - - - -**

A. - 3abi
B. 3ab
C. **- 3ab**D. 3abi

---

**Q.5 \begin{gathered} If A = \{ 1, 3, 5, 7, 9\} \hfill \ {\rm B} = \{ 2, 4, 6, 8, 10\} \hfill \ C = \{ 1, 2, 3, 4, 5, 6, 7, 8, 9, 10\} \hfill \ then what is A \cap B \cap C = ? \hfill \\ \end{gathered}**

**A. {}**
**B.** \{ 2, 4, 6, 8, 10\}
**C.** \{ 1, 3, 5, 7, 9\}
**D.** \{ 1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}

---

**Q.6 Cardinality of the Intersection of A={0,1,2} and B={1,2,3} is …….**

A. 3
B. 1
C. 4
D. **2**

---

**Q.7 Set of all the persons with horns on their heads is an example of…….**

- **A. Empty set**
- **B.** Infinite Set
- **C.** Finite Set with single element
- **D.** Universal Set

---

**Q.8 The\,\,value\,\,of\,\,z - \overline z \,\,is\,\, - - - - - .**

- A. 1
- B. 0
- C. **complex\,\,number**
- D. rea{{ l}} \,\,number

**Q.9 Cardinalities of sets {} and {{}} are……**

- **A. Not equal**
- **B.** Equal

---

**Q.10 Domain of function: f(x)=Sqrt(x-1) is all Real numbers ---------.**

- **A. x>1**

**(note: Sqrt means square root of.)**

- **B.** x>1 and x=1
- **C.** x<1
- **D.** x<1 and x=1

---